Full-Band Speech Enhancement via Low-Band Analysis and Observed High-Band Query Refinement

December 2026

Department of Artificial Intelligence

Graduate School, Sogang University

Woo Seok Kim

Full-Band Speech Enhancement via Low-Band Analysis and Observed High-Band Query Refinement

Advisor : Prof. Hyung-Min Park

A thesis submitted in partial fulfillment of the requirements for the degree of Master in Artificial Intelligence

December 2026

Department of Artificial Intelligence

Graduate School, Sogang University

Woo Seok Kim

논 문 인 준 서

김우석의 공학석사 학위논문을 인준함

2026 년 12 월

주심 김 지 환 ㉧

부심 장 두 성 ㉧

부심 박 형 민 ㉧

Contents

List of Figures

List of Tables

Abstract

This thesis presents TF-Rehancer V5, a time-frequency speech enhancement architecture for direct full-band processing. The model treats high-frequency components in 48 kHz speech as observed but noisy spectral evidence rather than as missing information to be generated from a band-limited input. TF-Rehancer uses sampling-frequency-independent STFT analysis, power-law compressed complex spectral conditioning, low-band encoder-guided refinement of observed high-band query features, and local complex time-frequency filtering over the original noisy STFT.

The evaluation is organized around two roles. First, a 16 kHz VoiceBank+DEMAND benchmark is used to compare quality and efficiency against lightweight speech enhancement references. In this setting, TF-Rehancer achieves PESQ-WB 3.334, DNSMOS SIG 3.405, DNSMOS BAK 4.042, and DNSMOS OVRL 3.129 with 0.470M parameters and 1.890G model-side MACs. Compared with the locally evaluated FastEnhancer-M checkpoint, TF-Rehancer improves PESQ-WB and DNSMOS SIG/BAK/OVRL with lower MACs, while FastEnhancer-M remains stronger in DNSMOS P.808, SCOREQ, SI-SDR, STOI, and ESTOI. The result therefore supports a competitive quality-efficiency trade-off rather than uniform dominance.

Second, 48 kHz experiments are used as controlled full-band ablations. Models are trained with VCTK clean speech mixed with DNS full-band noise and evaluated on the VoiceBank+DEMAND 48 kHz test set. These experiments are not used as external full-band system rankings. The 48 kHz ablations show that replacing complex TF filtering with direct spectrum mapping degrades full-reference, spectral, and non-intrusive metrics. They also show that decoder cross-attention changes the trade-off between perceptual scores and spectral fidelity, and that a 5 kHz low-band cutoff provides a balanced operating point under the controlled ablation protocol.

Chapter 1

Introduction

Many established speech enhancement benchmarks and baselines are built around conventional 16 kHz wideband settings. In this setting, the upper-band spectral components contained in 48 kHz full-band speech are outside the model input. Direct 48 kHz full-band speech enhancement therefore presents a different operating condition: the model receives a noisy full-band signal, and the upper-band spectrum is available as noisy evidence to be enhanced. Although full-band enhancement has been explored in prior work, same-protocol references for direct 48 kHz full-band SE remain less established than conventional wideband evaluation.

Speech bandwidth extension and speech super-resolution target a different information condition. These tasks aim to reconstruct missing or degraded high-frequency content from band-limited inputs [1, 2], whereas 48 kHz full-band SE starts from a noisy full-band observation. Therefore, the key issue is not to synthesize the upper band after wideband enhancement, but to suppress noise over the full observed spectrum while preserving useful spectral evidence. This distinction motivates a direct full-band enhancement approach in which the high-frequency region is treated as part of the noisy observation rather than as a missing component to be generated afterward.

Practical full-band systems such as PercepNet and DeepFilterNet2 have shown that full-band enhancement can be made feasible under strict computational constraints [3, 4]. PercepNet uses perceptually motivated critical-band features and comb filtering, while DeepFilterNet2 combines ERB-domain enhancement with deep filtering. Such perceptual-band front ends trade bin-level spectral resolution for efficiency, which motivates a complementary TF-domain design that uses a learned frequency representation and estimates the enhanced spectrum through complex filtering over the original noisy STFT.

In this work, we propose TF-Rehancer V5, a refinement-oriented time-frequency architecture for full-band speech enhancement. The core idea is to encode low-band speech-structure evidence and use the observed high-band spectrum as decoder-side query information. Through cross-band interaction, TF-Rehancer V5 refines the high-band query and the subsequent full-band representation. The enhanced spectrum is then estimated by complex filtering over the original noisy STFT, rather than by directly generating a clean spectrum from a latent representation. This design allows the model to exploit cross-band speech structure while keeping the final estimation tied to the observed noisy signal.

We evaluate TF-Rehancer V5 with a 16 kHz VoiceBank+DEMAND comparison and 48 kHz controlled full-band ablations. The 16 kHz experiment examines the quality-efficiency trade-off against lightweight speech enhancement references. The 48 kHz experiments analyze the architectural components of TF-Rehancer under full-band processing and are

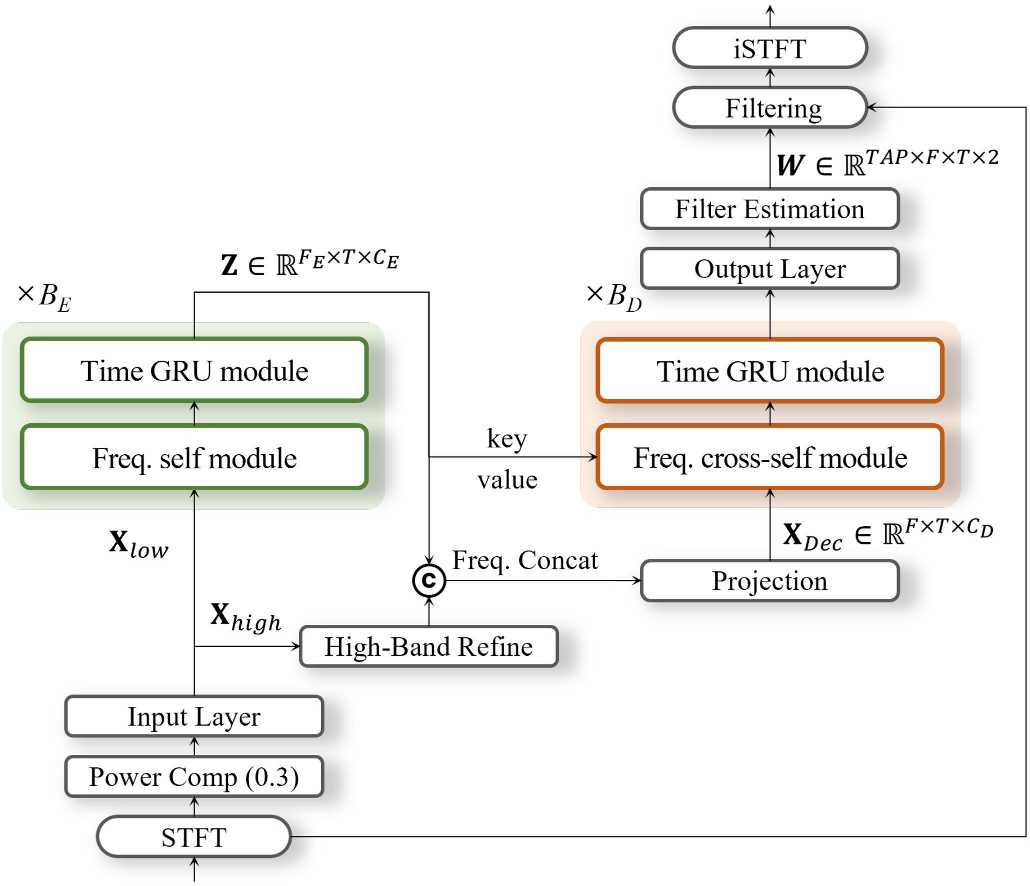

Figure 1.1: Conceptual overview of TF-Rehancer V5. The model analyzes low-band speech evidence and refines an observed high-band query before local complex filtering and waveform reconstruction.

not used as external full-band system rankings.

The contributions of this work are summarized as follows.

1. We formulate 48 kHz full-band SE as observation-preserving refinement, where upper-band components are treated as noisy spectral evidence rather than missing content to be generated after wideband processing.

2. We propose TF-Rehancer V5, a TF-domain architecture that combines low-band

speech-structure evidence, decoder-side observed high-band query features, cross-band interaction, and complex filtering over the original noisy STFT.

3. We provide a 16 kHz quality-efficiency comparison and controlled 48 kHz ablations that analyze complex filtering, decoder cross-attention, and the low-band cutoff without using them as external full-band system rankings.

The remainder of this thesis is organized as follows. Chapter 2 reviews related work in wideband enhancement, bandwidth extension, and practical full-band enhancement. Chapter 3 describes the TF-Rehancer V5 architecture and training objective. Chapter 4 presents the experimental setup, benchmark results, and ablation analysis. Chapter 5 summarizes the thesis and discusses limitations and future work.

Chapter 2

Related Work

2.1 Wideband Speech Enhancement, Bandwidth Extension, and the Observed-Band Gap

Neural speech enhancement (SE) has evolved from time-frequency mask estimation to phase-aware, complex-domain, time-domain, and modern TF-domain modeling. Early systems commonly estimated real-valued masks in the STFT domain, while later methods introduced phase-sensitive masks, complex ratio masks, and complex-valued networks to better handle phase-related distortion. Time-domain systems replaced the fixed STFT representation with learned analysis and synthesis bases, and recent TF-domain models further improved enhancement quality by explicitly modeling temporal and spectral dependencies [5, 6, 7, 8, 9, 10, 11]. However, many reported comparisons remain tied to 16 kHz or protocol-specific settings, making direct same-protocol 48 kHz full-band comparison less standardized.

The motivation for full-band SE is also supported by perceptual studies on high-frequency speech information. Extended high-frequency cues above the conventional wideband range have been shown to contribute to speech perception in noise and to the perception of fricative-related cues [12, 13, 14]. These findings motivate retaining and refining upper-band observations when the full-band spectrum is already present in the input.

Speech bandwidth extension and speech super-resolution address a different information condition from direct full-band SE. These tasks aim to reconstruct or supplement missing high-frequency content from band-limited inputs, often targeting a higher output sampling rate or wider output bandwidth [1, 2, 15]. In contrast, 48 kHz full-band SE starts from a noisy full-band observation, where the upper-band spectrum is already present but corrupted by noise. Therefore, full-band SE requires noise suppression and spectral refinement over the observed full-band signal, rather than only high-band reconstruction from a band-limited input.

2.2 Practical Full-Band and Frequency-Structured Speech Enhancement

Practical full-band SE has been studied mainly under real-time and low-complexity constraints. PercepNet enhances 48 kHz speech using perceptually motivated critical-band features and comb filtering, while DeepFilterNet2 combines ERB-domain enhancement with deep filtering for low-complexity full-band enhancement [3, 4]. These systems are important references because they show that 48 kHz full-band enhancement can be made practical under strict computational constraints. Such perceptual or ERB-band representations trade bin-level spectral resolution for efficiency, motivating TF-domain designs that preserve explicit frequency-bin evidence while still controlling computational cost.

Frequency-structured modeling has also become a major design axis in modern SE and source separation. Subband and band-split models process spectra through grouped frequency regions, while TF-domain dual-path models separate temporal and spectral modeling to improve efficiency and representation power. BSRNN and BS-RoFormer show that band-wise modeling is effective for high-resolution audio modeling, and TF-Locoformer develops an RNN-free TF-domain dual-path architecture with local modeling by convolution [16, 17, 11]. Lightweight systems such as GTCRN, LiSenNet, and FastEnhancer further show that compact recurrent, subband, and frequency-time modeling can improve the performance-efficiency trade-off in real-time SE [18, 19, 20].

Many band-split and lightweight models emphasize compact subband grouping or efficient within-band/global sequence modeling. TF-Rehancer V5 follows a cross-band refinement principle: low-band speech-structure evidence is encoded and used to refine observed high-band query features. This design targets direct 48 kHz full-band SE, where the high-frequency region is not missing but available as noisy spectral evidence.

2.3 Query-Based Refinement and Filtering-Based Enhancement

Query-based restoration provides another relevant direction. TF-Restormer introduces an asymmetric encoder-decoder architecture for speech restoration with decoupled input and output sampling rates. Its encoder focuses on the observed input bandwidth, while the decoder restores target frequency regions through frequency extension queries [21]. This mechanism is well matched to bandwidth restoration, where decoder-side queries can serve as slots for frequency regions that are absent from the input.

TF-Restormer also uses a sampling-frequency-independent STFT (SFI-STFT) formulation for variable-rate restoration. Instead of fixing the STFT window and hop in samples, SFI-STFT fixes their physical durations, so the sample-level window and hop scale with the sampling rate. With a 40 ms analysis window and a 20 ms hop, the temporal frame geometry remains consistent across common sampling rates, while the number of frequency bins scales with the output bandwidth. This provides a useful frontend principle for comparing 16 kHz and 48 kHz TF-domain models without changing the temporal analysis scale.

Direct 48 kHz full-band SE has a different information condition. The high-frequency region is already present in the noisy input, so a learned missing-band placeholder is less natural than a query derived from the observed high-band spectrum. TF-Rehancer reinterprets query refinement for full-band SE: the decoder-side high-band query is derived from the observed high-band spectrum and refined using low-band speech-structure evidence. This changes the role of query modeling from missing-band reconstruction to observed-band refinement.

Filtering-based enhancement is closely related to this observation-preserving view. Instead of directly mapping a latent representation to a clean spectrum, deep filtering estimates complex local filters and applies them to the noisy STFT [4]. Recent correlation-to-filter approaches such as SR-CorrNet provide related evidence that correlation-derived or structure-conditioned filters can be effective alternatives to direct spectral mapping [22]. TF-Rehancer follows this filtering-based principle by estimating the enhanced spectrum through complex filtering over the original noisy STFT.

TF-Rehancer V5 combines these two lines of work in the full-band SE setting. From query-based restoration, it adopts the idea that decoder-side query information can guide frequency-region refinement. From filtering-based enhancement, it adopts the principle that enhancement can be constrained through complex filtering over the observed noisy STFT. Its key information source is the observed input itself: the high-band query is not a learned missing-band token, but an observed high-band representation refined by low-band speech-structure evidence.

Chapter 3

Proposed Method

3.1 Problem Formulation

Let $y \in \mathbb{R}^L$ and $x \in \mathbb{R}^L$ denote the noisy and clean speech waveforms, respectively. In single-channel speech enhancement, the noisy signal is modeled as

$$y = x + n, \tag{3.1}$$

where n denotes additive noise. Applying the short-time Fourier transform (STFT) to y and x gives the noisy and clean complex spectra

$$Y, X \in \mathbb{C}^{F \times T}, \tag{3.2}$$

where F and T denote the number of frequency bins and time frames. The target of full-band speech enhancement is to estimate a clean full-band spectrum $\hat{X}$ from the noisy full-band observation Y, followed by inverse STFT reconstruction of the enhanced waveform $\hat{x}$.

TF-Rehancer V5 follows an observation-preserving filtering formulation. The neural network predicts a complex local time-frequency filter W, and the final spectrum is obtained by applying this filter to a normalized linear noisy STFT reference Y^{ref}:

$$W = \mathcal{G}_\theta(\Phi(Y)), \qquad \hat{X} = W \otimes Y^{\mathrm{ref}}. \tag{3.3}$$

Here, $\mathcal{G}_\theta$ denotes TF-Rehancer V5, $\Phi(Y)$ is the neural conditioning feature, and $\otimes$ denotes local complex filtering over a time-frequency neighborhood. The compressed feature $\Phi(Y)$ is used for conditioning, while the output operator is applied to the original linear noisy spectrum. Therefore, the model is not trained as an unconstrained direct generator of $\hat{X}$; it estimates the filter coefficients used to transform the observed noisy spectrum.

For a 48 kHz full-band input, the upper band is already present in Y, although it is corrupted by noise. TF-Rehancer V5 therefore treats the high-frequency region as observed noisy evidence rather than as missing information. The model encodes low-band speech structure, derives decoder-side high-band query features from the observed high band, and refines the full-band representation before complex filtering.

3.2 SFI-STFT Conditioning and Input/Output Layers

TF-Rehancer V5 adopts the sampling-frequency-independent STFT (SFI-STFT) principle used in TF-Restormer [21]. Let f_s denote the sampling rate, and let the analysis window and hop durations be $\tau_w = 0.04$ s and $\tau_h = 0.02$ s. The STFT parameters are set

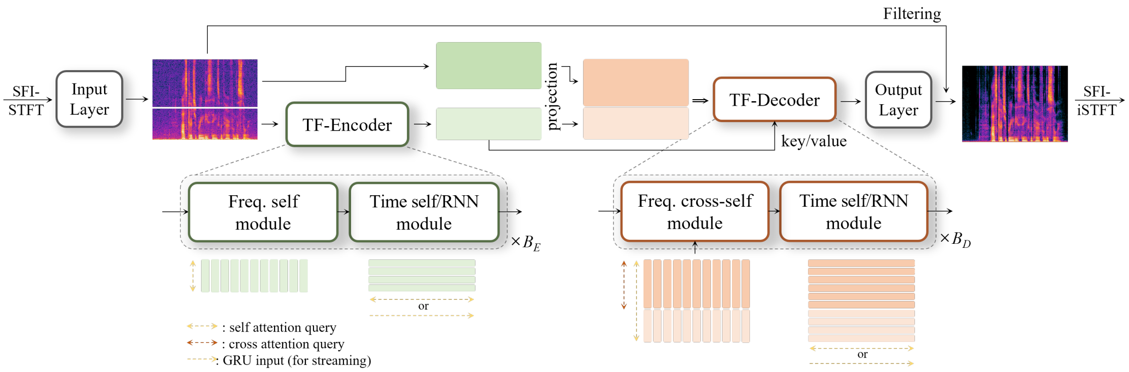

Figure 3.1: Overall TF-Rehancer V5 architecture. The shared input layer produces an embedded full-band representation, the low-band region is analyzed by the TF encoder, the observed high-band region is carried as decoder-side query evidence, and the TF decoder refines the full-band representation before the output layer estimates complex TF filters.

by physical duration:

$$N(f_s) = \tau_w f_s, \qquad H(f_s) = \tau_h f_s, \qquad F(f_s) = \frac{N(f_s)}{2} + 1. \tag{3.4}$$

Thus, the 16 kHz configuration uses $N = 640$, $H = 320$, and $F = 321$, while the 48 kHz configuration uses $N = 1920$, $H = 960$, and $F = 961$. This keeps the temporal frame geometry fixed across sampling rates and lets the frequency-bin count scale with bandwidth. Unlike TF-Restormer, which uses decoupled input-output rates for restoration and super-resolution, TF-Rehancer uses the same SFI-STFT parameterization for the noisy input, clean target, and inverse-STFT reconstruction in matched-rate 16 kHz and 48 kHz speech enhancement.

Figure 3.1 summarizes the canonical TF-Rehancer V5 architecture used in this thesis. The frontend constructs a power-law compressed conditioning feature while preserving a linear noisy STFT reference for the final filtering step. The shared input layer embeds the full-band STFT and reduces only the frequency resolution. A fixed 5 kHz split then routes low-band tokens to the TF encoder and high-band tokens to the decoder-side observed-query path. The encoder uses four low-band time-frequency blocks to model speech-structure evidence. The decoder uses two cross-band time-frequency blocks to refine the high-band query and the joint full-band representation. Finally, FastFilterUpsample restores the original STFT frequency resolution, and the output layer estimates a local 3×3 complex filter applied to Y^{ref}.

The neural conditioning feature is constructed from a power-law compressed complex STFT. Given $\gamma = 0.3$, the compressed spectrum is

$$C_\gamma(Y) = Y \left(|Y|^2 + \epsilon\right)^{(\gamma-1)/2}. \tag{3.5}$$

The input feature concatenates compressed real, compressed imaginary, and compressed magnitude channels:

$$\Phi(Y) = \left[\mathrm{Re}(C_\gamma(Y)), \mathrm{Im}(C_\gamma(Y)), |C_\gamma(Y)|\right]. \tag{3.6}$$

Power-law compression stabilizes the dynamic range of spectral amplitudes for neural processing. The uncompressed noisy STFT reference Y^{ref} is kept separately for the final filtering operation.

The feature $\Phi(Y)$ is passed through a shared frequency-strided input stem:

$$H = \mathrm{Stem}(\Phi(Y)). \tag{3.7}$$

The stem consists of Conv2D($3 \to 48$), batch normalization, and a SiLU activation. The convolution uses kernel size $(4, 1)$, stride $(2, 1)$, and padding $(2, 0)$, so it reduces only the frequency resolution and preserves the frame rate. This shared stem is used for both 16 kHz and 48 kHz configurations. After the stem, the embedded frequency resolution is $F_{\text{emb}} = 161$ at 16 kHz and $F_{\text{emb}} = 481$ at 48 kHz.

On the output side, the decoder representation is restored to the original STFT frequency resolution before estimating local complex filter coefficients. Thus, both input-side embedding and output-side filtering are described as local convolutional transformations around an explicit STFT reference, rather than as unconstrained waveform generation.

3.3 Observed High-Band Query and Low-Band Analysis

After the shared stem, TF-Rehancer V5 splits the embedded frequency axis using a 5 kHz cutoff:

$$(H_{\text{low}}, H_{\text{high}}) = \text{Split}_{5\text{kHz}}(H). \tag{3.8}$$

At 16 kHz, this gives $F_{\text{low}} = 101$ and $F_{\text{high}} = 60$. At 48 kHz, it gives $F_{\text{low}} = 101$ and $F_{\text{high}} = 380$. Thus, the same low-band speech-structure region is encoded at both sampling rates, while the 48 kHz model keeps a much larger observed high-band region in the decoder path.

The high-band branch is not a learned empty query. It is derived from the observed high-band stem feature:

$$Q_h = H_{\text{high}} + \text{Adapter}_h(H_{\text{high}}). \tag{3.9}$$

The adapter is a residual local convolutional block with SiLU activation and layer normal-

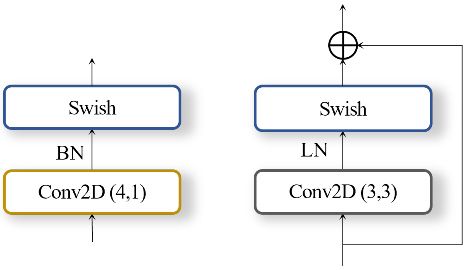

Figure 3.2: Local convolutional layer notation used for compactly describing the input-side, output-side, and high-band-refinement layer family. The exact channel width, stride, and normalization for each use are specified in the text.

ization, using the local layer family summarized in Fig. 3.2. This block aligns the observed high-band representation with the decoder query space before cross-band refinement. It does not perform decoder refinement by itself; it prepares the high-band tokens for the decoder.

The low-band encoder analyzes H_{low} with four stacked blocks:

$$Z = \text{Encoder}(H_{\text{low}}). \tag{3.10}$$

Each encoder block contains an efficient frequency feed-forward module, frequency-axis multi-head self-attention with rotary positional encoding, another frequency feed-forward module, and a FastGRU2 temporal module. The encoder does not use a frequency projection layer or a pre-encoder sinusoidal positional encoding. Its role is to model speech structure from the low-band region, where harmonic and formant-related evidence is relatively dense.

3.4 Cross-Band Decoder Refinement

The decoder first concatenates the low-band encoder output and the adapted high-band query along the embedded frequency axis:

$$D_0 = \mathrm{LN}\left(\mathrm{Linear}\left([Z; Q_h]_F\right)\right). \tag{3.11}$$

The projection maps the concatenated 48-channel representation to the 24-channel decoder width. This projection is a separate alignment stage between the encoder/query interface and the decoder blocks.

TF-Rehancer V5 uses two decoder blocks. In each block, multi-head cross-attention updates only the high-band tokens by using the encoded low-band representation as key and value:

$$\tilde{D}_{\mathrm{high}} = D_{\mathrm{high}} + \mathrm{MHCA}(D_{\mathrm{high}}, Z, Z). \tag{3.12}$$

The updated high-band tokens are concatenated back with the low-band decoder tokens:

$$\tilde{D} = [D_{\mathrm{low}}; \tilde{D}_{\mathrm{high}}]_F. \tag{3.13}$$

Subsequent frequency-axis self-attention, frequency feed-forward processing, and FastGRU2 temporal modeling operate on the full embedded sequence. Cross-attention therefore provides directed low-to-high guidance, while the following full-sequence modules refine the joint full-band representation.

This design separates three roles. The high-band adapter constructs an observed high-band query from H_{high}. The concat-projection layer aligns the low-band encoder output

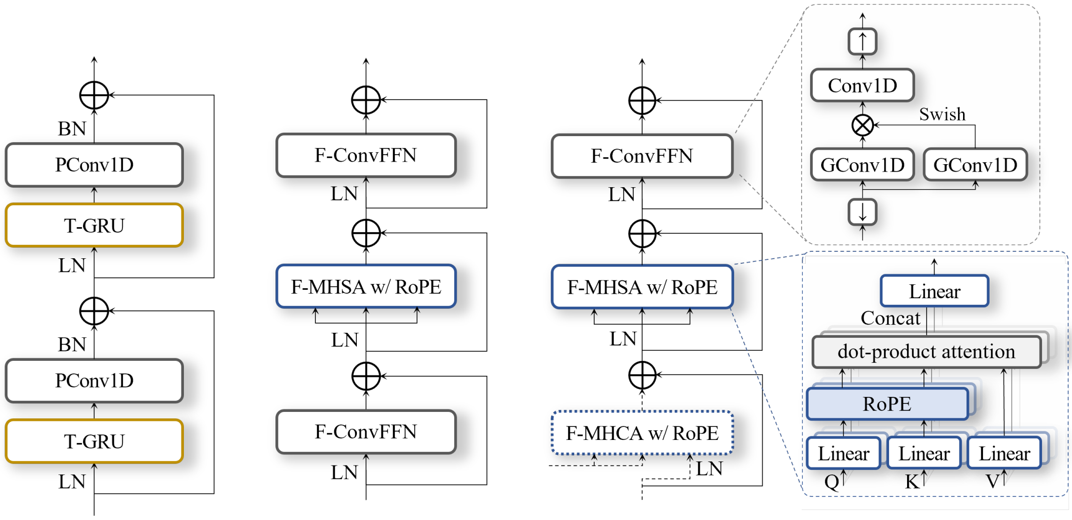

Figure 3.3: TF-Rehancer encoder and decoder core blocks. The temporal module uses residual PConv1D–T-GRU stages, the encoder frequency module combines F-ConvFFN and F-MHSA with RoPE, and the decoder frequency module adds F-MHCA with RoPE before full-sequence self-attention and convolutional feed-forward processing.

and high-band query into the decoder width. The decoder blocks then perform cross-band refinement and full-sequence time-frequency modeling.

3.5 Full-Resolution Complex TF Filtering

The decoder output is still at embedded frequency resolution. Before filter estimation, TF-Rehancer V5 restores it to the original STFT frequency resolution using FastFilterUpsample:

$$D_{\text{full}} = \text{FastFilterUpsample}(D_{\text{emb}}) \in \mathbb{R}^{B \times F \times T \times C_d}. \tag{3.14}$$

This module uses a pointwise convolution followed by transposed convolution along the frequency axis. The restored full-resolution feature is then passed to a partial-convolution filter head and a local 3×3 time-frequency convolution. The output-side convolutional operations follow the compact local-layer notation in Fig. 3.2, while the explicit filtering operation is described below.

For each time-frequency bin, the head predicts nine complex taps corresponding to

$$\Delta f \in \{-1, 0, 1\}, \qquad \Delta t \in \{-1, 0, 1\}. \tag{3.15}$$

The raw head output has 27 channels: one magnitude scale, one real component, and one imaginary component for each of the nine taps. A softplus activation constrains the magnitude scale to be positive, and tanh activations bound the real and imaginary components:

$$M = \mathrm{softplus}(A_m), \qquad R, I = \tanh(A_{ri}), \tag{3.16}$$

$$W_{\Delta f,\Delta t,f,t} = M_{\Delta f,\Delta t,f,t}\left(R_{\Delta f,\Delta t,f,t} + jI_{\Delta f,\Delta t,f,t}\right). \tag{3.17}$$

The enhanced spectrum is computed as

$$\hat{X}_{f,t} = \sum_{\Delta f=-1}^{1} \sum_{\Delta t=-1}^{1} W_{\Delta f,\Delta t,f,t} Y^{\mathrm{ref}}_{f+\Delta f,t+\Delta t}. \tag{3.18}$$

Boundary positions are padded before local neighborhoods are gathered. This operation is a complex filtering operation, not a correlation estimate. The coefficients W are the predicted filter weights used to construct $\hat{X}$; they are conditioned on the decoder feature and applied to the observed noisy STFT reference.

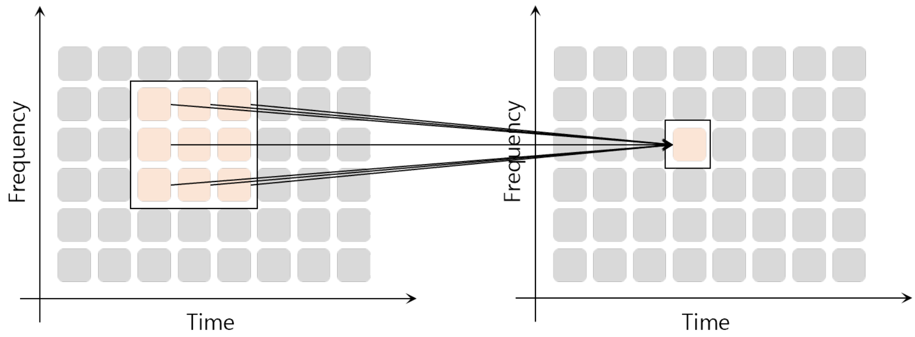

Figure 3.4: Local 3×3 complex TF filtering. For each output bin, TF-Rehancer V5 predicts complex coefficients over a local time-frequency neighborhood and applies them to the corresponding noisy STFT samples.

3.6 Training Objective

TF-Rehancer V5 is trained with a corrected FastEnhancer-style composite enhancement loss. The spectral terms are computed on gamma-compressed complex spectra. For a complex spectrum X, the compressed spectrum is

$$X_c = |X|^\gamma \exp(j\angle X), \qquad \gamma = 0.3, \tag{3.19}$$

with an ϵ-stabilized magnitude in implementation. The total loss is

$$\mathcal{L} = \lambda_{\text{mag}}\mathcal{L}_{\text{mag}} + \lambda_{\text{cplx}}\mathcal{L}_{\text{cplx}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}} + \lambda_{\text{wav}}\mathcal{L}_{\text{wav}} + \lambda_{\text{pesq}}\mathcal{L}_{\text{pesq}}. \tag{3.20}$$

The compressed magnitude loss is

$$\mathcal{L}_{\text{mag}} = \left\| |\hat{X}_c| - |X_c| \right\|_2^2. \tag{3.21}$$

The compressed complex loss is

$$\mathcal{L}_{\text{cplx}} = \left\| \text{Re}(\hat{X}_c) - \text{Re}(X_c) \right\|_2^2 + \left\| \text{Im}(\hat{X}_c) - \text{Im}(X_c) \right\|_2^2 . \tag{3.22}$$

The consistency loss compares the compressed STFT of the reconstructed waveform with the clean compressed spectrum:

$$\mathcal{L}_{\text{cons}} = \|\text{STFT}(\hat{x})_c - X_c\|_2^2 . \tag{3.23}$$

The waveform-domain term is

$$\mathcal{L}_{\text{wav}} = \|\hat{x} - x\|_1. \tag{3.24}$$

A weak PESQ auxiliary term is also used:

$$\mathcal{L}_{\text{pesq}} = \text{PESQ-Loss}(\hat{x}, x). \tag{3.25}$$

For 48 kHz training, this auxiliary term is evaluated after resampling the waveforms to 16 kHz. The loss weights are

$$\lambda_{\text{mag}} = 0.3, \quad \lambda_{\text{cplx}} = 0.2, \quad \lambda_{\text{cons}} = 0.3, \quad \lambda_{\text{wav}} = 0.2, \quad \lambda_{\text{pesq}} = 0.001. \tag{3.26}$$

The objective combines spectral reconstruction, waveform fidelity, STFT consistency after waveform reconstruction, and a small perceptual proxy term. The primary supervision remains the compressed spectral and waveform-domain enhancement losses.

Chapter 4

Experiments

4.1 Experimental Setup

4.1.1 Datasets and Protocols

We evaluate TF-Rehancer using a 16 kHz wideband benchmark and a 48 kHz controlled ablation setting. The 16 kHz setting uses the official VoiceBank+DEMAND (VBD) corpus to compare the proposed model with lightweight speech enhancement baselines. The 48 kHz setting uses VCTK clean speech mixed with DNS full-band noise to analyze the architectural components of TF-Rehancer under full-band processing.

For the 16 kHz comparison, we use the official VBD paired speech enhancement corpus. The official training partition contains 28 speakers and is split into 10,413 training utterances and 1,159 validation utterances using a fixed random seed. Evaluation is performed on the official 824-utterance test set, which contains unseen speakers.

For the 48 kHz ablation study, noisy mixtures are synthesized from VCTK [23] clean speech and DNS [24] full-band noise. Each model is trained for 20 epochs and evaluated on the VBD 48 kHz test set. These experiments are intended for controlled component analysis, not for ranking against external full-band systems.

4.1.2 Model Configuration

This chapter evaluates the final TF-Rehancer architecture described in Chapter 3. The main design principle is low-band encoder-guided refinement of observed high-band features in the full-band decoder. The model concentrates heavy encoding on the low-band representation, keeps observed high-band features in the decoder path, refines the decoder representation using the low-band encoder output, and estimates the enhanced spectrum through complex TF filtering over the original noisy STFT.

All TF-Rehancer experiments use a 40 ms analysis window and a 20 ms frame shift. With this setting, both 16 kHz and 48 kHz configurations have a frequency spacing of 25 Hz. The default low-band cutoff is 5 kHz, corresponding to 201 full-resolution frequency bins and 101 embedded low-band tokens after the frequency-strided stem. Thus, the low-band encoder length is nearly fixed across 16 kHz and 48 kHz configurations, while the number of observed high-band decoder tokens increases in the 48 kHz setting.

Model complexity is reported using the number of parameters and model-side multiply-accumulate operations (MACs). Unless otherwise stated, MACs exclude STFT and inverse STFT operations.

4.1.3 Evaluation Metrics

For locally trained TF-Rehancer runs, evaluation uses the checkpoint with the lowest validation loss. The default VoiceBank+DEMAND evaluation set contains 824 utterances.

We use both full-reference and non-intrusive metrics. DNSMOS P.808 and DNSMOS P.835 SIG/BAK/OVRL are computed after resampling the waveform to 16 kHz. SCOREQ is computed in full-reference natural-speech mode after 16 kHz resampling. PESQ-WB, STOI, and ESTOI are also computed after 16 kHz resampling. Unless otherwise specified, SI-SDR is computed at the task sampling rate.

For the 48 kHz analysis, we additionally report full-band spectral metrics because many standard speech enhancement metrics are computed after 16 kHz resampling and do not directly evaluate the upper-frequency region. We report full-band LSD, MCD, and SI-SDR for 48 kHz outputs. DNSMOS SIG/BAK/OVRL refers to DNSMOS P.835, while CSIG/CBAK/COVL refers to composite full-reference metrics.

4.2 16 kHz Speech Enhancement and Efficiency

We first evaluate the 16 kHz configuration on a standard wideband benchmark to compare enhancement quality and computational cost.

Table 4.1 reports 16 kHz results without a separate source column. BSRNN follows the FastEnhancer paper [20]. FastEnhancer-S and -M use the released checkpoints and local evaluation. TF-Rehancer uses the same evaluator on the 824-utterance VBD test set. Therefore, the FastEnhancer and TF-Rehancer rows provide the closest same-evaluator comparison, whereas the BSRNN row serves as benchmark context.

Relative to the locally evaluated FastEnhancer-M checkpoint, TF-Rehancer improves

Method	Params↓	MACs↓	PESQ-WB↑	STOI↑	ESTOI↑	SI-SDR↑	SCOREQ↓	DNSMOS(P.835)			DNSMOS
								SIG↑	BAK↑	OVL↑	(P.808)↑
BSRNN	0.334M	**0.25G**	3.06	0.942	0.855	18.90	0.303	3.30	4.00	3.07	3.44
FastEnhancer-M	0.492M	2.90G	3.24	0.950	0.873	19.40	**0.243**	3.39	4.02	3.11	3.48
TF-Rehancer (on)	0.470M	1.89G	3.33	0.950	0.872	19.44	0.244	3.41	4.04	3.13	**3.49**
TF-Rehancer (off)	0.615M	2.00G	**3.41**	**0.953**	**0.876**	**19.68**	0.249	**3.42**	**4.07**	**3.16**	**3.49**

Table 4.1: VoiceBank+DEMAND 16 kHz test results. MACs are model-side estimates and exclude STFT/iSTFT.

PESQ-WB and DNSMOS SIG/BAK/OVRL. It also uses lower model-side MACs and a similar number of parameters. FastEnhancer-M remains better in DNSMOS P.808, SCOREQ, SI-SDR, STOI, and ESTOI. Relative to FastEnhancer-S, TF-Rehancer improves PESQ-WB, DNSMOS SIG/BAK/OVRL, SI-SDR, STOI, and ESTOI, but requires more computation. Therefore, the 16 kHz result supports a competitive quality-efficiency trade-off rather than uniform dominance.

The BSRNN comparison should not be interpreted as a same-evaluator superiority claim. It indicates that TF-Rehancer reaches a competitive region relative to a published lightweight reference, while using more parameters and MACs than the reported BSRNN configuration.

4.3 Full-Band Architecture Ablations

The direct mapping head is weaker than complex TF filtering across the reported metric families. This suggests that filtering-based output estimation is more reliable than direct spectrum mapping in the 48 kHz ablation setting. The w/o decoder MHCA variant shows a mixed trend. It improves PESQ, STOI, CSIG, COVL, and DNSMOS SIG, but degrades SI-SDR, LSD, MCD, and DNSMOS BAK/OVRL. Thus, decoder MHCA should

Method	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SI-SDR↑	LSD↓	MCD↓	DNSMOS(P.835)↑ SIG	BAK	OVL	DNSMOS (P.808)↑
TF-Rehancer	**3.29**	0.947	4.41	**3.60**	3.88	**16.84**	**0.72**	**2.27**	3.38	**3.98**	**3.07**	**3.46**
w/o Decoder MHCA	**3.29**	**0.948**	**4.44**	3.59	**3.90**	16.38	0.74	2.28	**3.39**	3.96	**3.07**	**3.46**
w/ Direct Mapping	3.05	0.940	4.24	3.40	3.66	14.90	0.78	2.52	3.35	3.96	3.03	3.44

Table 4.2: 48 kHz architecture ablation on the VoiceBank+DEMAND 48 kHz test set. All models are trained for 20 epochs using VCTK clean speech and DNS full-band noise and evaluated on the VoiceBank+DEMAND 48 kHz test set. These results are intended for controlled component analysis, not for ranking against external full-band systems.

be interpreted as changing the trade-off between perceptual scores and spectral fidelity, rather than as a uniformly beneficial module.

4.4 Low-Band Cutoff Sensitivity

Cutoff	MACs↓	PESQ↑	STOI↑	CSIG↑	CBAK↑	COVL↑	SI-SDR↑	LSD↓	MCD↓	DNSMOS(P.835) SIG↑	BAK↑	OVL↑	DNSMOS (P.808)↑
8 kHz	3.81G	**3.31**	0.946	**4.42**	**3.60**	**3.90**	16.50	0.75	2.38	3.39	3.98	**3.08**	3.46
7 kHz	3.54G	3.29	0.947	4.40	3.57	3.88	16.11	0.75	2.35	3.39	3.96	3.07	3.45
6 kHz	3.28G	3.29	0.947	4.42	3.60	3.89	16.75	0.72	**2.26**	3.39	3.97	3.07	**3.46**
5 kHz	3.02G	3.29	0.947	4.41	3.60	3.88	**16.84**	**0.72**	2.27	3.38	**3.98**	3.07	**3.46**
4 kHz	2.76G	3.27	0.947	4.41	3.56	3.87	16.29	0.73	2.31	3.39	3.97	3.07	3.45
3 kHz	**2.50G**	3.23	0.946	4.40	3.51	3.84	15.41	0.78	2.38	**3.40**	3.92	3.06	3.45

Table 4.3: Low-band cutoff ablation. All rows use the 48 kHz 20-epoch setting with VCTK clean speech and DNS full-band noise. Parameter counts are identical across cutoff settings and are omitted. MACs are model-side estimates.

The cutoff sweep shows a trade-off among computation, perceptual quality, and spectral fidelity. Larger cutoffs increase computation and improve some perceptual scores, whereas smaller cutoffs reduce MACs but can degrade full-band and high-band spectral fidelity. Under the 48 kHz ablation setting, we use 5 kHz as the default cutoff because it provides a balanced operating point between computational cost, SI-SDR, and high-band spectral metrics.

Chapter 5

Conclusion

5.1 Summary

This thesis presented TF-Rehancer, a time-frequency-domain speech enhancement architecture for efficient full-band processing. The main idea is low-band encoder-guided full-band decoder refinement: TF-Rehancer concentrates heavier modeling on low-band speech structure, keeps observed upper-band evidence in the decoder path, and estimates the enhanced spectrum through complex time-frequency filtering over the original noisy STFT.

The experiments show that TF-Rehancer provides a competitive quality-efficiency trade-off on the 16 kHz VoiceBank+DEMAND benchmark. Compared with the locally evaluated FastEnhancer-M checkpoint, TF-Rehancer improves PESQ-WB and DNSMOS SIG/BAK/OVRL with fewer model-side MACs and a similar number of parameters, while FastEnhancer-M remains stronger in DNSMOS P.808, SCOREQ, SI-SDR, STOI, and

ESTOI. In the controlled 48 kHz ablation setting, direct spectrum mapping degrades full-reference, spectral, and non-intrusive metrics compared with complex TF filtering, suggesting that filtering-based output estimation is more reliable for the proposed architecture. The decoder MHCA ablation shows a mixed trend, indicating that cross-attention changes the trade-off between perceptual scores and spectral fidelity rather than providing a uniformly beneficial gain. The low-band cutoff analysis further shows that 5 kHz provides a balanced operating point between computational cost, waveform fidelity, and high-band spectral behavior.

Overall, these results support low-band encoder-guided full-band decoder refinement as an efficient design direction for full-band speech enhancement. The architecture preserves observed upper-band information, avoids heavy full-band encoding, and uses full-resolution complex TF filtering to produce the enhanced spectrum.

5.2 Limitations and Future Work

The 48 kHz experiments in this work are designed for controlled component analysis, not for ranking against external full-band speech enhancement systems. A protocol-aligned 48 kHz comparison with strong full-band baselines remains necessary to establish broader performance claims. Future work should therefore evaluate TF-Rehancer and competing full-band systems under a shared training and evaluation protocol, including aligned metric computation and runtime measurement.

Several architectural questions also remain open. The present ablation supports complex TF filtering as the output estimator and characterizes the effect of decoder MHCA, but it does not fully separate the contribution of the lightweight high-band adaptation from the subsequent decoder refinement. A focused ablation that removes or replaces the local high-band adapter while preserving the observed high-band embedding would clarify this component. In addition, the low-band cutoff was selected as a balanced operating point under the 48 kHz ablation setting; broader datasets, noise conditions, and sampling rates may require adaptive or data-dependent cutoff strategies.

Finally, practical deployment requires further validation beyond model-side MACs. Future work should include real-time runtime profiling, latency analysis, and evaluation on device-oriented scenarios. Extension to multi-channel or reverberant conditions is another possible direction.

Bibliography

[1] H. Liu, X. Liu, Q. Kong, Q. Tian, Y. Zhao, D. Wang, C. Huang, and Y. Wang, "Voicefixer: A unified framework for high-fidelity speech restoration," in *Proc. Interspeech*, pp. 4232–4236, 2022.

[2] P. Andreev, A. Alanov, O. Ivanov, and D. Vetrov, "Hifi++: A unified framework for bandwidth extension and speech enhancement," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2023.

[3] J.-M. Valin, U. Isik, N. Phansalkar, R. Giri, K. Helwani, and A. Krishnaswamy, "A perceptually-motivated approach for low-complexity, real-time enhancement of fullband speech," in *Proc. Interspeech*, pp. 2482–2486, 2020.

[4] H. Schröter, A. Maier, A. N. Escalante-B., and T. Rosenkranz, "Deepfilternet2: Towards real-time speech enhancement on embedded devices for full-band audio," in *Proc. International Workshop on Acoustic Signal Enhancement (IWAENC)*, pp. 1–5, 2022.

[5] H. Erdogan, J. R. Hershey, S. Watanabe, and J. L. Roux, "Phase-sensitive and recognition-boosted speech separation using deep recurrent neural networks," in

Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 708–712, 2015.

[6] D. S. Williamson, Y. Wang, and D. Wang, "Complex ratio masking for monaural speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 24, no. 3, pp. 483–492, 2016.

[7] Y. Hu, Y. Liu, S. Lv, M. Xing, S. Zhang, Y. Fu, J. Wu, B. Zhang, and L. Xie, "Dccrn: Deep complex convolution recurrent network for phase-aware speech enhancement," in *Proc. Interspeech*, pp. 2472–2476, 2020.

[8] Y. Luo and N. Mesgarani, "Conv-tasnet: Surpassing ideal time-frequency magnitude masking for speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 27, no. 8, pp. 1256–1266, 2019.

[9] R. Cao, S. Abdulatif, and B. Yang, "Cmgan: Conformer-based metric gan for speech enhancement," in *Proc. Interspeech*, pp. 936–940, 2022.

[10] Y.-X. Lu, Y. Ai, and Z.-H. Ling, "Mp-senet: A speech enhancement model with parallel denoising of magnitude and phase spectra," in *Proc. Interspeech*, pp. 3834–3838, 2023.

[11] K. Saijo, G. Wichern, F. G. Germain, Z. Pan, and J. L. Roux, "Tf-locoformer: Transformer with local modeling by convolution for speech separation and enhancement," in *Proc. International Workshop on Acoustic Signal Enhancement (IWAENC)*, pp. 205–209, 2024.

[12] P. G. Stelmachowicz, A. L. Pittman, B. M. Hoover, and D. E. Lewis, "Effect of

stimulus bandwidth on the perception of /s/ in normal- and hearing-impaired children and adults," *Journal of the Acoustical Society of America*, vol. 110, no. 4, pp. 2183–2190, 2001.

[13] B. B. Monson, A. J. Lotto, and B. H. Story, "Detection of high-frequency energy level changes in speech and singing," *Journal of the Acoustical Society of America*, vol. 135, no. 1, pp. 400–406, 2014.

[14] B. C. J. Moore, "A review of the perceptual effects of hearing loss for frequencies above 3 khz," *International Journal of Audiology*, vol. 55, no. 12, pp. 707–714, 2016.

[15] H. Liu, K. Chen, Q. Tian, W. Wang, and M. D. Plumbley, "Audiosr: Versatile audio super-resolution at scale," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1076–1080, 2024.

[16] Y. Luo and J. Yu, "Music source separation with band-split rnn," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1893–1901, 2023.

[17] W.-T. Lu, J.-C. Wang, Q. Kong, and Y.-N. Hung, "Music source separation with band-split rope transformer," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 481–485, 2024.

[18] X. Rong, T. Sun, X. Zhang, Y. Hu, C. Zhu, and J. Lu, "Gtcrn: A speech enhancement model requiring ultralow computational resources," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 971–975, 2024.

[19] H. Yan, J. Zhang, C. Fan, Y. Zhou, and P. Liu, "Lisennet: Lightweight sub-band and

dual-path modeling for real-time speech enhancement," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2025.

[20] S. Ahn, J. Han, B. J. Woo, and N. S. Kim, "Fastenhancer: Speed-optimized streaming neural speech enhancement," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.

[21] U.-H. Shin, J. Ko, W. Jeong, and H.-M. Park, "Query-based asymmetric modeling with decoupled input-output rates for speech restoration," in *Proc. International Conference on Machine Learning (ICML)*, 2026. To appear.

[22] U.-H. Shin and H.-M. Park, "Asymmetric encoder-decoder based on time-frequency correlation for speech separation," *arXiv preprint arXiv:2603.29097*, 2026.

[23] C. Veaux, J. Yamagishi, and K. MacDonald, "Cstr vctk corpus: English multi-speaker corpus for cstr voice cloning toolkit," 2017.

[24] C. K. A. Reddy, V. Gopal, R. Cutler, E. Beyrami, R. Cheng, H. Dubey, S. Matusevych, R. Aichner, A. Aazami, S. Braun, P. Rana, S. Srinivasan, and J. Gehrke, "The interspeech 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results," in *Proc. Interspeech*, pp. 2492–2496, 2020.